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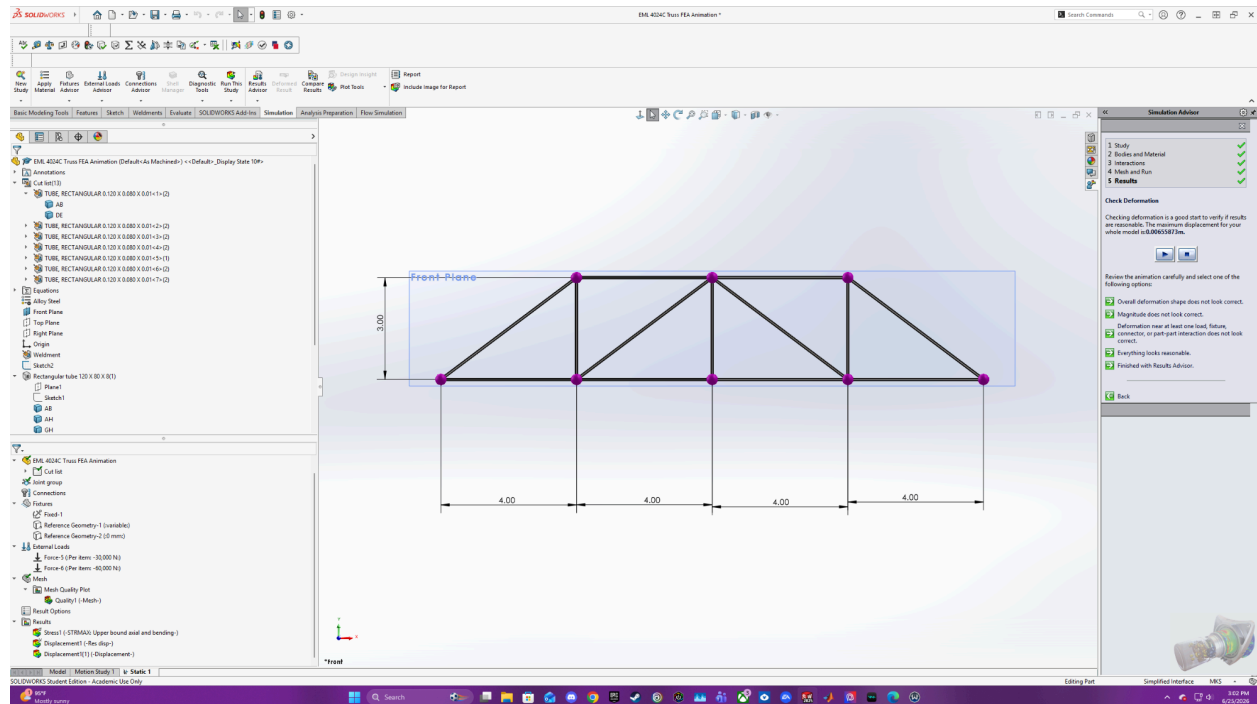
EML 4024C Engineering Design Practice

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Project 2 Truss FEA Animation Report

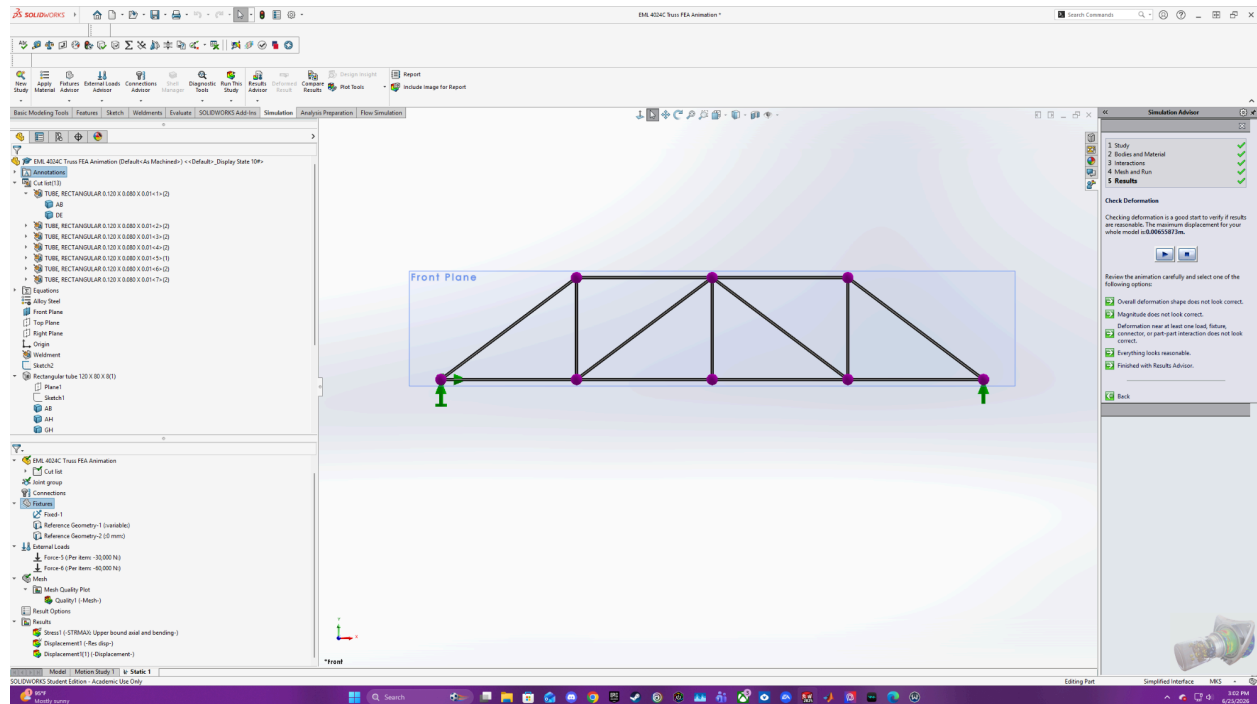
The purpose of this project was to design and analyze a truss structure using SOLIDWORKS Simulation. The truss was created using weldment structural members and tested using finite element analysis to determine the stress, displacement, and internal member forces caused by external loading. The goal of the simulation was to evaluate how the truss behaved under the applied forces and to determine whether the structure could support the loading without excessive stress or displacement.

The truss was modeled in SOLIDWORKS using a two dimensional sketch on the Front Plane. The geometry was then converted into a three dimensional truss using structural members. The selected profile was an ISO rectangular tube with dimensions of 120 millimeters by 80 millimeters by 8 millimeters. The material assigned to the truss was alloy steel. The final truss design used four horizontal spans of four meters each and a vertical height of three meters. Since the diagonal members used a three meter by four meter triangle, each diagonal member had a length of five meters.



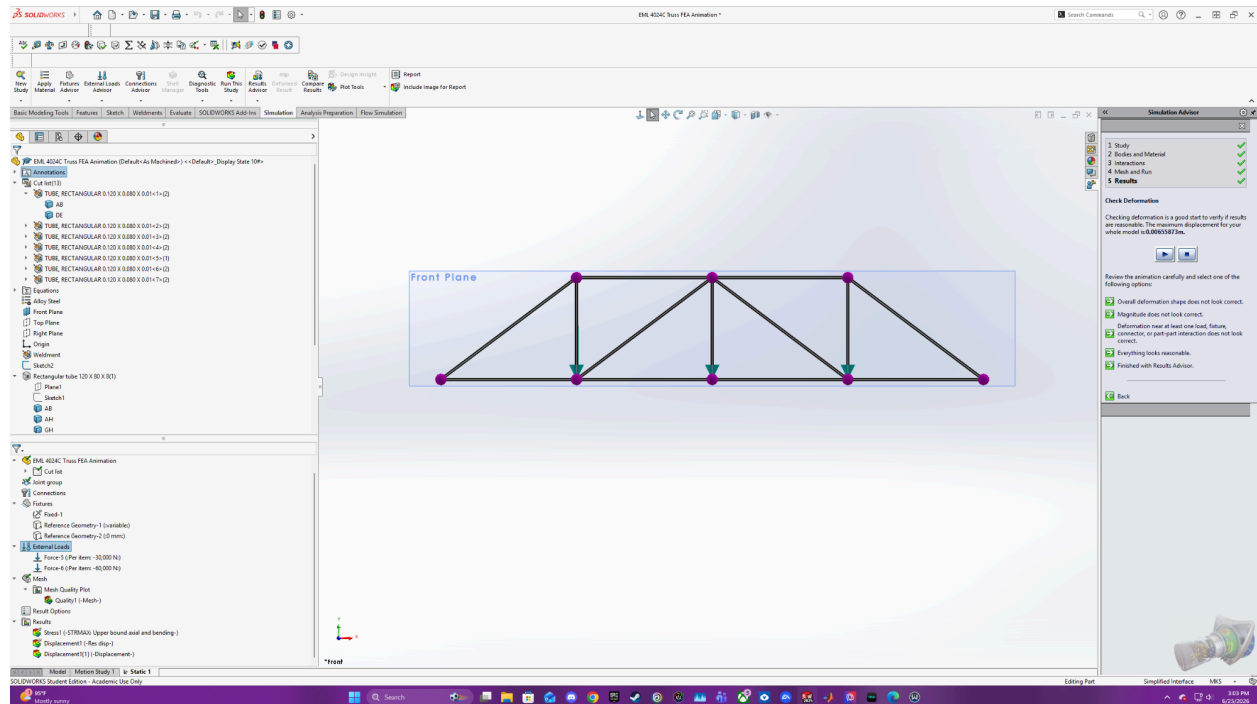
This figure shows the final truss geometry with the selected dimensions. The truss had four four meter horizontal sections and a three meter height. The diagonal members were five meters long because they were based on the 3-4-5 triangle relationship.

The truss was supported using one fixed support and one roller support. The fixed support was placed at the left joint of the truss. This support prevented the structure from moving in all directions at that joint. The roller support was placed at the right joint of the truss. This support prevented vertical motion while still allowing horizontal movement. This support setup allowed the truss to behave like a simply supported structure.



This figure shows the support conditions used in the SOLIDWORKS Simulation study. The left joint was fixed, and the right joint was modeled as a roller support. These supports were important because they controlled how the truss reacted to the external forces.

The external forces were applied to the bottom joints of the truss. A downward force of 30 kilonewtons was applied at joint H, a downward force of 60 kilonewtons was applied at joint G, and a downward force of 30 kilonewtons was applied at joint F.

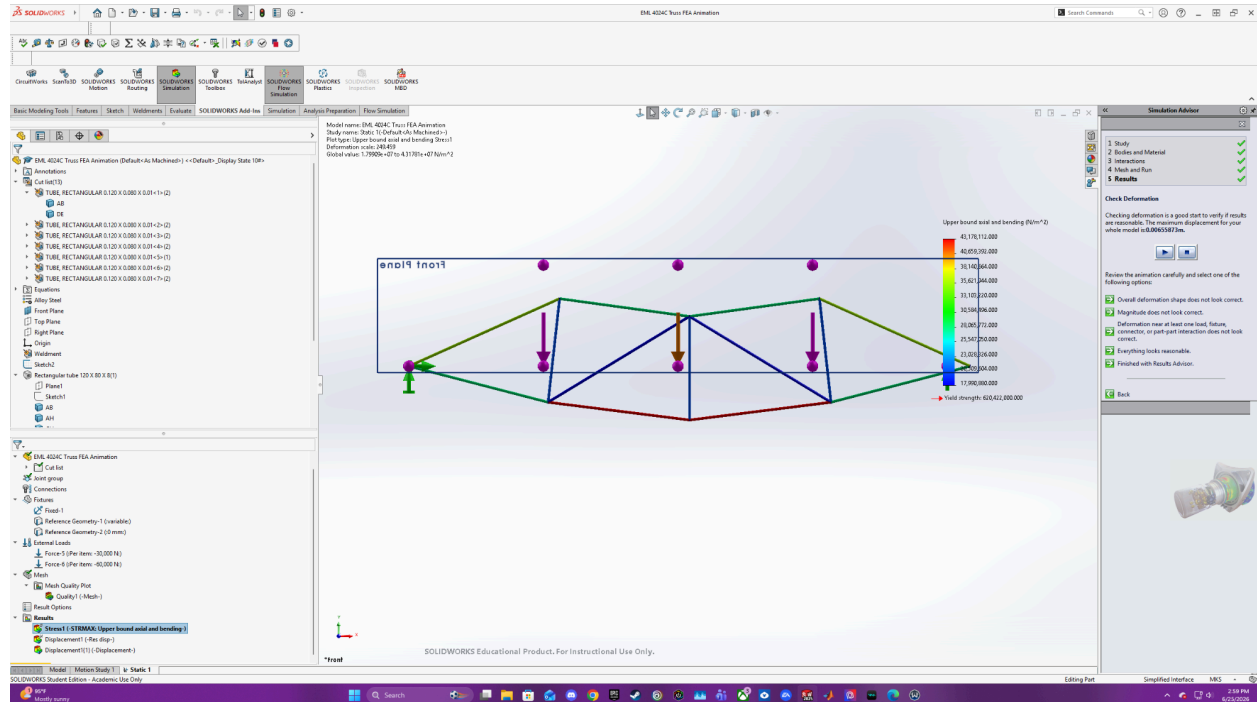


This figure shows the external forces applied to the truss. The two outer downward loads were 30 kN each, and the middle downward load was 60 kN. These forces caused the truss to bend downward and created internal tension and compression forces in the members.

After the supports and external forces were applied, the structural members were converted from beam elements to truss elements. This was done because a truss member is assumed to carry axial force only. This means each member is mainly in either tension or compression. The static study was then run in SOLIDWORKS Simulation to calculate the stress, displacement, and member force results.

The stress plot showed how the applied loads affected the internal stress in the truss members. The maximum stress from the simulation was approximately 43.18 MPa. The yield strength of the alloy steel material was approximately 620.42 MPa. Since the maximum stress

was lower than the yield strength, the truss did not reach material failure under the applied loading.



This figure shows the stress distribution of the truss. The maximum stress was approximately 43.18 MPa, which was much lower than the alloy steel yield strength of approximately 620.42 MPa. This means the truss stayed within a safe stress range during the simulation.

The factor of safety was calculated by comparing the yield strength to the maximum stress.

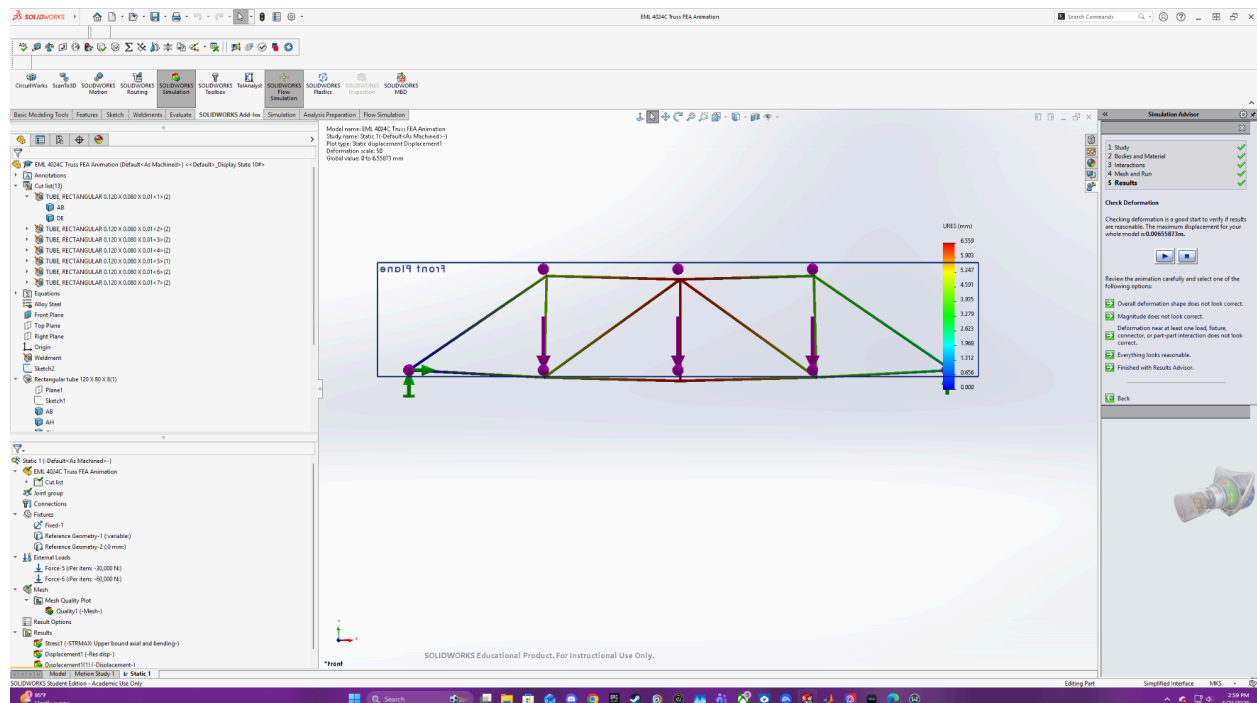
Factor of safety equals yield strength divided by maximum stress.

Factor of safety equals 620.42 MPa divided by 43.18 MPa.

Factor of safety equals approximately 14.37.

This factor of safety shows that the truss had a large margin between the maximum simulated stress and the material yield strength. A higher factor of safety means the structure is less likely to fail under the applied loading. In this case, the truss was able to carry the applied forces without reaching the yield limit of the material.

The displacement plot showed how much the truss moved under the applied loading. The maximum displacement was approximately 6.559 millimeters. The minimum displacement was 0.000 millimeters, which occurred at the constrained support locations. The maximum displacement occurred near the loaded region of the truss, where the downward forces caused the structure to deflect.



This figure shows the displacement distribution of the truss. The maximum displacement

was approximately 6.559 millimeters. This means the truss moved downward slightly under the applied forces, but the displacement was small compared to the overall length of the structure.

The displacement result is important because a structure can have low stress but still move too much. If the displacement is too large, the structure may not be useful even if it does not break. In this simulation, the maximum displacement was small compared to the 16 meter total length of the truss. This suggests that the truss was stiff enough for the applied loading condition.

The internal forces in the truss members were also calculated by hand and compared with the SOLIDWORKS Simulation results. The hand calculation was done using static equilibrium and the method of joints. The support reactions were calculated first. Since the loading was symmetric, the left and right vertical reactions were both 60 kN. The horizontal reaction was 0 kN because there were no horizontal external loads.

The hand calculated member forces showed that member AH was in 80 kN of tension, member HG was in 120 kN of tension, member GF was in 120 kN of tension, and member FE was in 80 kN of tension. These results show that the bottom chord members were mainly being pulled apart by the applied loading.

The upper and outer diagonal members were mainly in compression. Member BA was in 100 kN of compression, member CB was in 80 kN of compression, member DC was in 80 kN of compression, and member ED was in 100 kN of compression. These members were being pushed together as the truss resisted the downward loads.

The remaining web members also had a combination of tension and compression.

Member BH was in 60 kN of tension, member HC was in 50 kN of compression, member CG was in 60 kN of tension, member CF was in 50 kN of compression, and member FD was in 60 kN of tension. This means the vertical members carried tensile forces, while the main diagonal web members carried compressive forces.

Beam Name	Element	End	Axial (N)	Shear1 (N)	Shear2 (N)	Moment 1 (N.m)	Moment 2 (N.m)	Torque (N.m)
Beam-1(CD)	---	---	80,000	0	0	0	0	0
Beam-2(AB)	---	---	1.00E+05	0	0	0	0	0
Beam-3(GH)	---	---	-1.20E+05	0	0	0	0	0
Beam-4(DE)	---	---	-40,000	0	0	0	0	0
Beam-5(BH)	---	---	-40,000	0	0	0	0	0
Beam-6(CH)	---	---	80,000	0	0	0	0	0
Beam-7(EF)	---	---	-40,000	0	0	0	0	0
Beam-8(CF)	---	---	50,000	0	0	0	0	0
Beam-9(CD)	---	---	-40,000	0	0	0	0	0
Beam-10(AH)	---	---	-40,000	0	0	0	0	0
Beam-11(FG)	---	---	-1.20E+05	0	0	0	0	0
Beam-12(DE)	---	---	1.00E+05	0	0	0	0	0
Beam-13(BC)	---	---	80,000	0	0	0	0	0

I have inserted the link to the spreadsheet into the image for easier access.

This figure shows the member force results from SOLIDWORKS Simulation. These simulation results were compared to the hand calculation results to check whether the model behaved as expected. The values should be similar because both the hand calculation and the simulation are based on static equilibrium. Any differences between the results may be caused by rounding, the sign convention used by SOLIDWORKS, or the way the software labels tension

and compression.

Finite element analysis is important for engineering design because it allows engineers to study a structure before it is physically built. Instead of building a real truss and testing it first, the model can be tested in SOLIDWORKS to predict stress, displacement, and internal forces. This saves time and materials because design problems can be found earlier in the design process.

FEA is also useful because it shows where the most important areas of the structure are located. The stress plot shows which members experience the highest stress, and the displacement plot shows where the truss moves the most. These results help engineers decide whether a design needs to be changed or strengthened. In this project, the FEA results showed that the truss stayed below the material yield strength and had a small maximum displacement.

The displacement animation also helped show how the truss deformed under the applied loading. The animation exaggerated the movement of the truss so that the deformation shape could be seen more clearly. This made it easier to understand where the truss was bending and how the structure responded to the forces.

The video shows the displacement animation of the truss under loading. The animation shows the deformed shape of the truss as the downward forces cause the structure to deflect.

Overall, the simulation showed that the truss was able to carry the applied loading without excessive stress or displacement. The maximum stress was approximately 43.18 MPa, which was much lower than the alloy steel yield strength of approximately 620.42 MPa. The

maximum displacement was approximately 6.559 millimeters, which was small compared to the full truss length.

The hand calculation showed that the truss members experienced a combination of tension and compression forces. The bottom chord members were mainly in tension, while the top chord and some diagonal members were in compression. This matched the expected behavior of a simply supported truss under downward loading.

The truss was successfully modeled and analyzed in SOLIDWORKS Simulation. The results showed that the selected truss design was reasonable for the applied loads. The stress results showed that the structure stayed within the safe material range, and the displacement results showed that the structure did not deform excessively. Overall, the finite element analysis provided a clear way to evaluate the truss design and compare the simulation results with hand calculations.